Welcome!

R. Heller · Date: 2026-05-11 · Last changed: 2026-11-10

Welcome to **“Scientometrics in R: Bibliometrics, Network Science, and Research Evaluation”**.

This is a working guide to scientometrics in R for researchers, librarians, research administrators, and data scientists who want to study scholarly communication quantitatively. It walks the full pipeline — from pulling records out of OpenAlex, Crossref, and PubMed, through computing the standard indicators (h-index, MNCS, journal metrics), to building co-authorship and co-citation networks, applying text and topic models to scientific corpora, and packaging the results into reproducible reports and dashboards.

The book assumes a working knowledge of R at the tidyverse level and basic statistics. No prior scientometric experience is needed — the early chapters lay the conceptual and ethical foundations, and each later chapter starts with a concrete research question and ends with a callout on responsible use grounded in the Leiden Manifesto and DORA.

All examples run against free, open data; no Web of Science or Scopus subscription is required. Small sample datasets ship with the companion R package `scientometricsInR`; everything else is fetched live from public APIs. The book does not cover proprietary platforms (InCites, SciVal), manual screening workflows for systematic reviews, or qualitative sociology of science.

Open Source Repository

This book has been built using **{rmarkdown}** and **{bookdown}**. Formulas are rendered using MathJax. All source files are available on **GitHub** at https://github.com/CTTIR/scientometrics-in-r. You are free to fork, share, and reuse contents under the project's CC BY 4.0 license.

How To Use The Guide

- **Linear or lookup.** Read front to back, or use the Find Your Method navigator to jump to the chapter that answers your question.
- **Code.** Reusable helpers live in the companion R package `scientometricsInR` and in `R/` in the source repository; every chapter shows the full chunk it depends on.
- **Downloads.** The side panel lists whole-book and chapter files that are available in the current build.
- **Corrections.** Please file an issue at https://github.com/CTTIR/scientometrics-in-r/issues.

Contributing

Pull requests, corrections, and additions are welcome. See `CONTRIBUTING.md` for the workflow.

Citing this Guide

The suggested citation is:

> **Heller, R.** (2026). Scientometrics in R: Bibliometrics, Network Science, and Research Evaluation (Version 0.1.0). Self-published via GitHub Pages. https://cttir.github.io/scientometrics-in-r/.

Download the reference as BibTeX or .ris.